

WEEK 7 PROJECT SUMMARY

1. What was planned

The Week 6 Random Forest candidate was planned for systematic testing covering predictive performance, errors, segments, generalisation, threshold behaviour, input robustness and business relevance.

2. What was tested

The candidate was tested using held-out data, confusion matrices, classification metrics, error analysis, segment testing, training/test comparison, five-fold cross-validation, input robustness testing and threshold analysis.

The model was also compared against the Week 5 Logistic Regression baseline.

3. Main testing results

The notebook generated evidence for Accuracy, Precision, Recall, F1-score and ROC-AUC, together with false-positive and false-negative counts.

The actual values should be taken directly from the generated Week 7 tables.

4. Issues identified

Testing investigated prediction errors, generalisation behaviour, threshold sensitivity and input robustness.

5. Refinement

A controlled Random Forest refinement was introduced using ``max_depth=12``.

6. Retesting

The refined model was re-tested on the same held-out data and through five-fold cross-validation.

7. End-to-end contribution

The Data Science component provides a predictive No-Show risk signal that can support administrative appointment planning and patient support activities.

8. Remaining limitations

The model remains dependent on dataset quality, sample representativeness, threshold selection and appropriate human oversight.

9. Week 8

The remaining work is to complete final integration, confirm pipeline compatibility, consolidate validation evidence and prepare the final HealthConnect presentation.